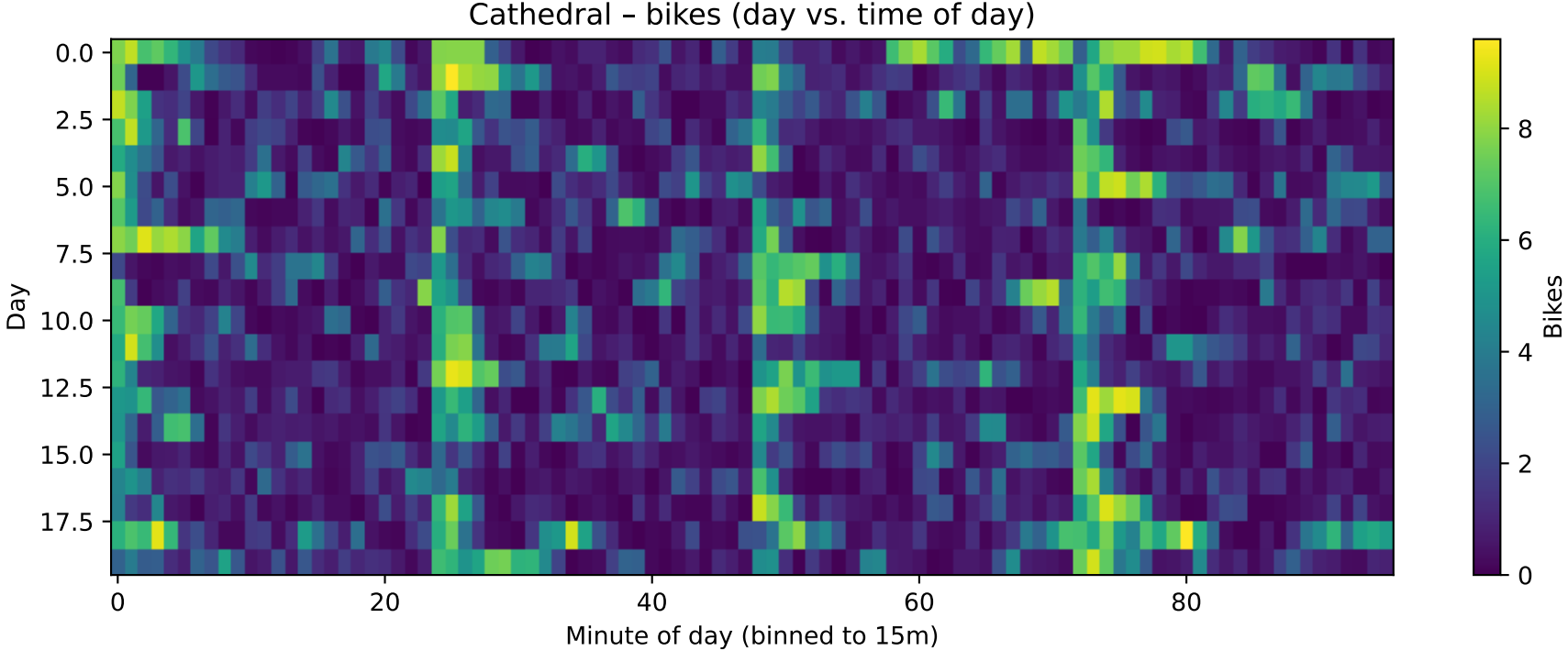
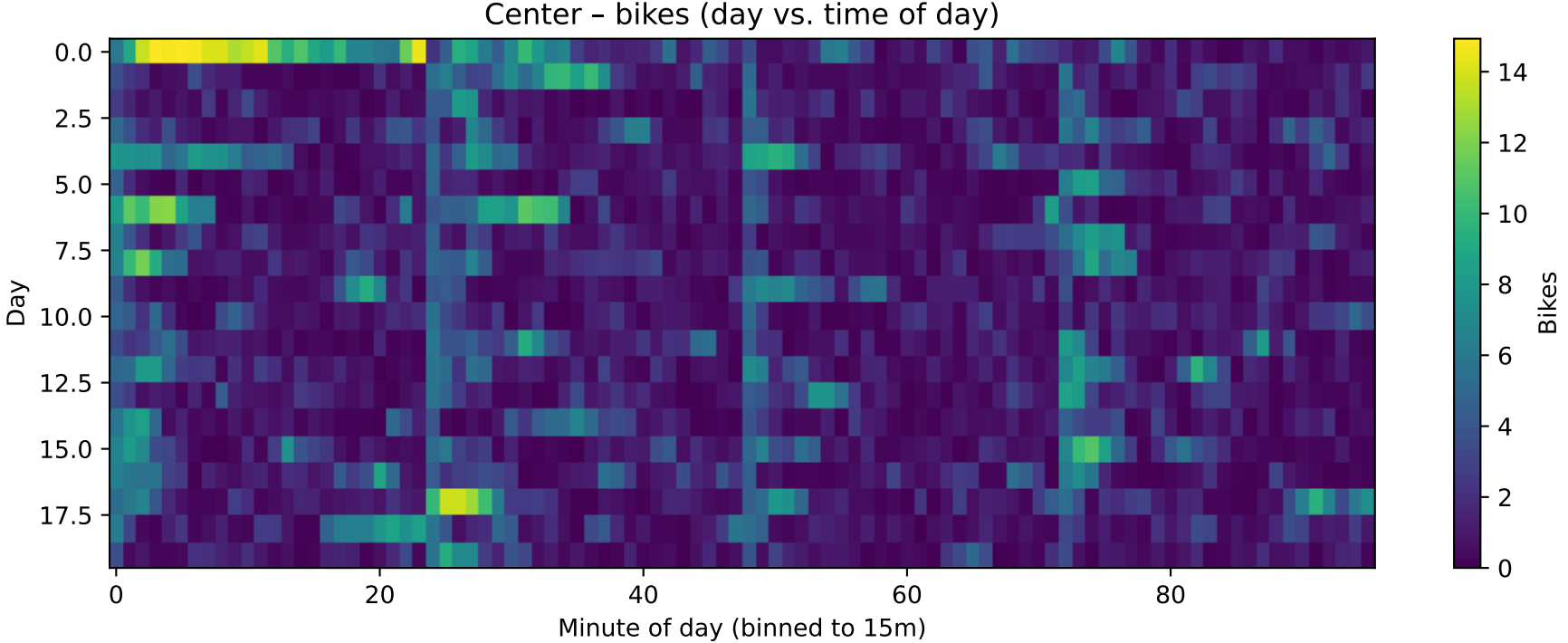
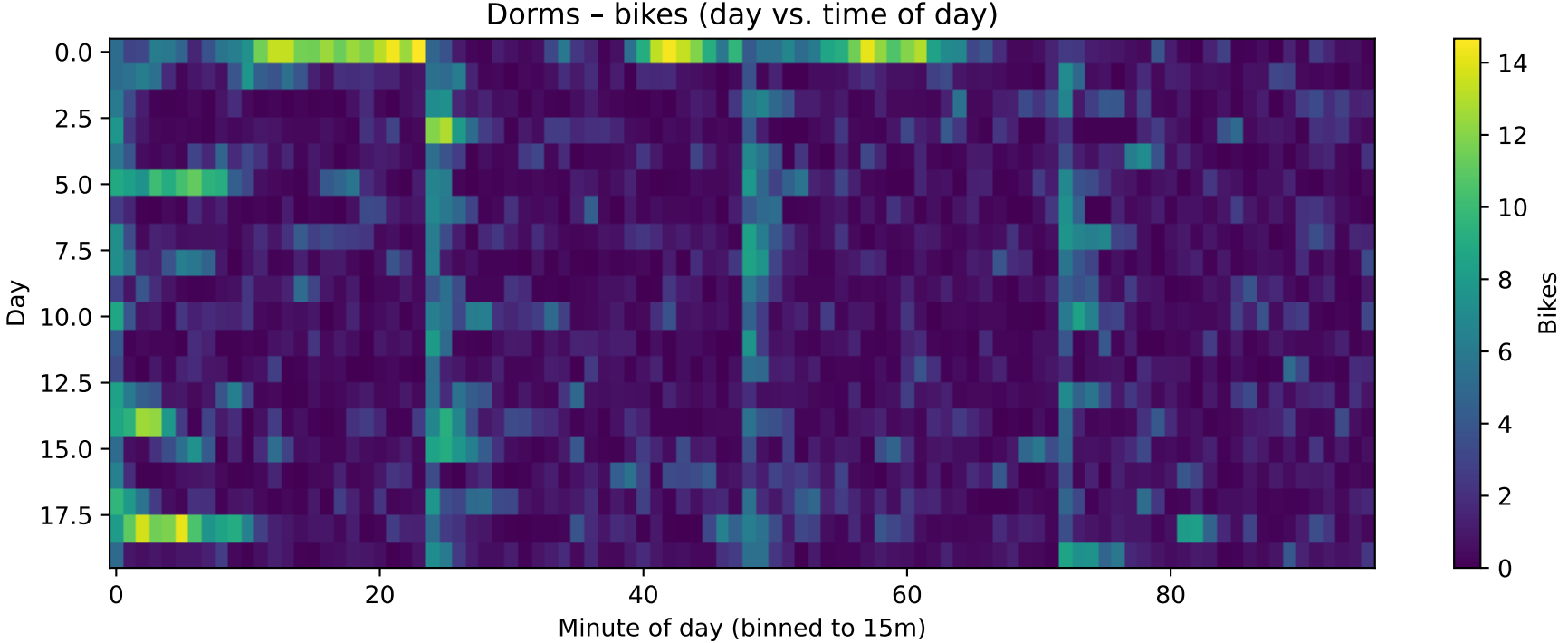
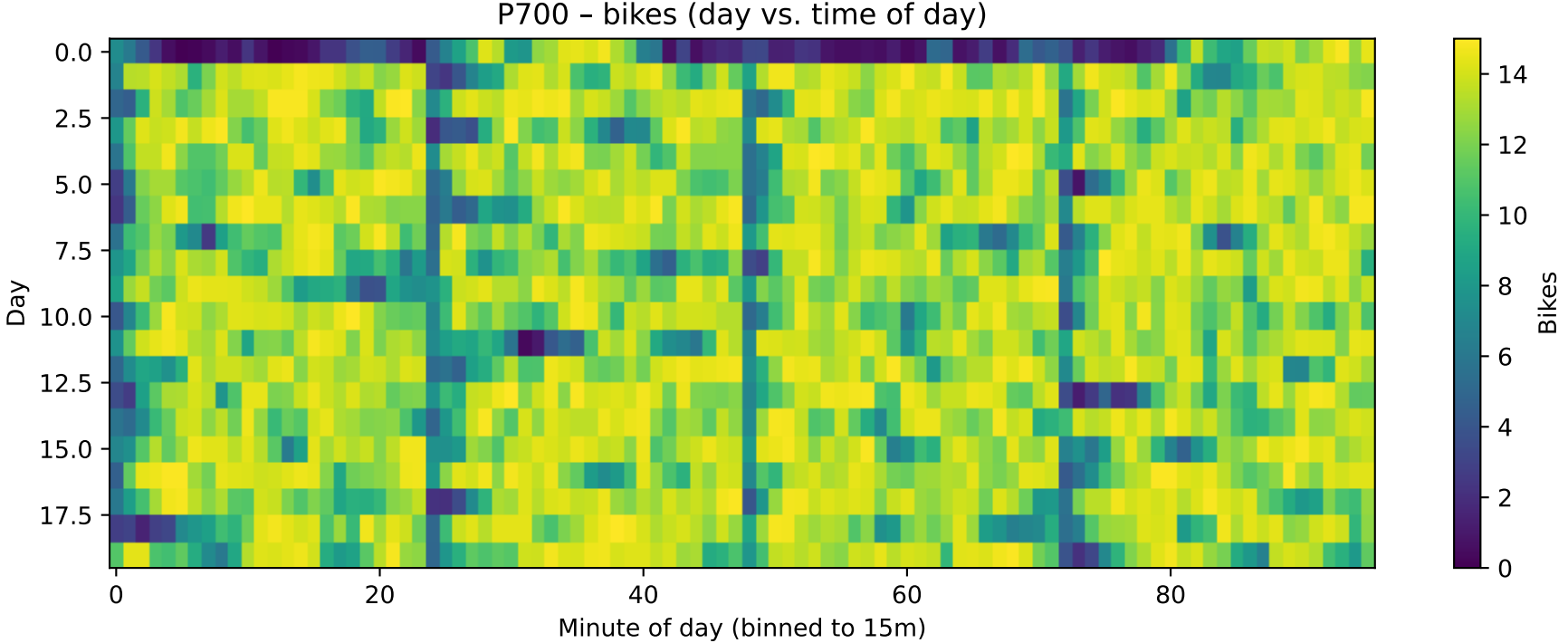
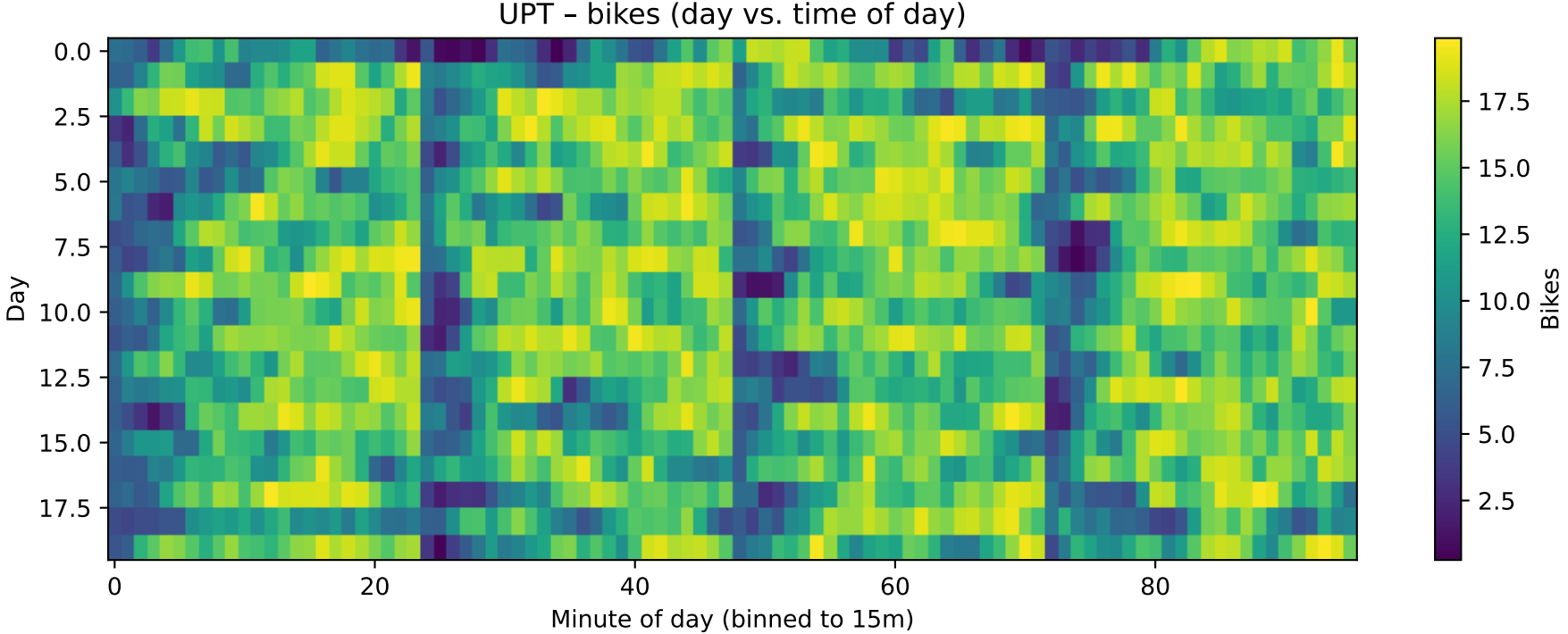
Simulation parameters:
- Duration: 20.0 days (28800 minutes)
- Rebalancing every: 360 minutes (0.25 days)
- Breakdown chance: 0.9%
- Repair time: 30 minutes
- Scrapped chance: 0.1%
- Trip time: 2 to 6 minutes
- Probabilities Matrix used: v1

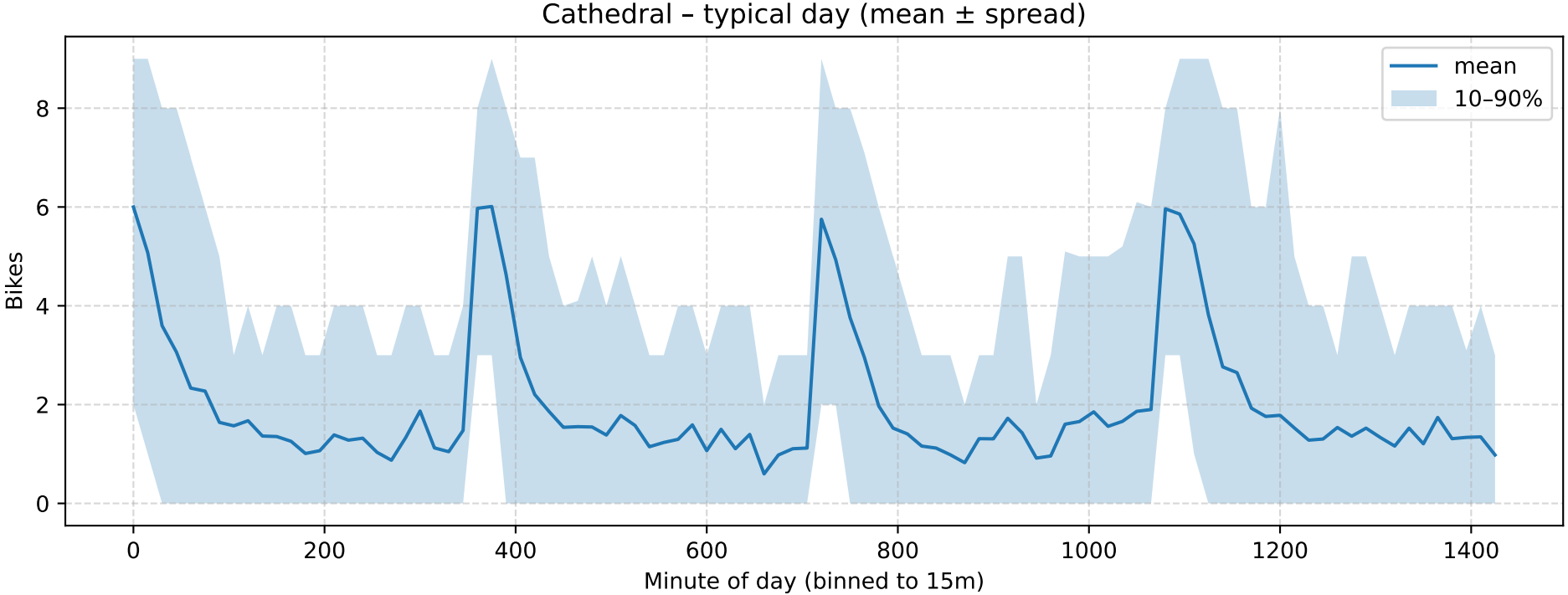

Cathedral – bikes (day vs. time of day)
Day
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
20
40
60
80
Minute of day (binned to 15m)
Bikes
0
2
4
6
8

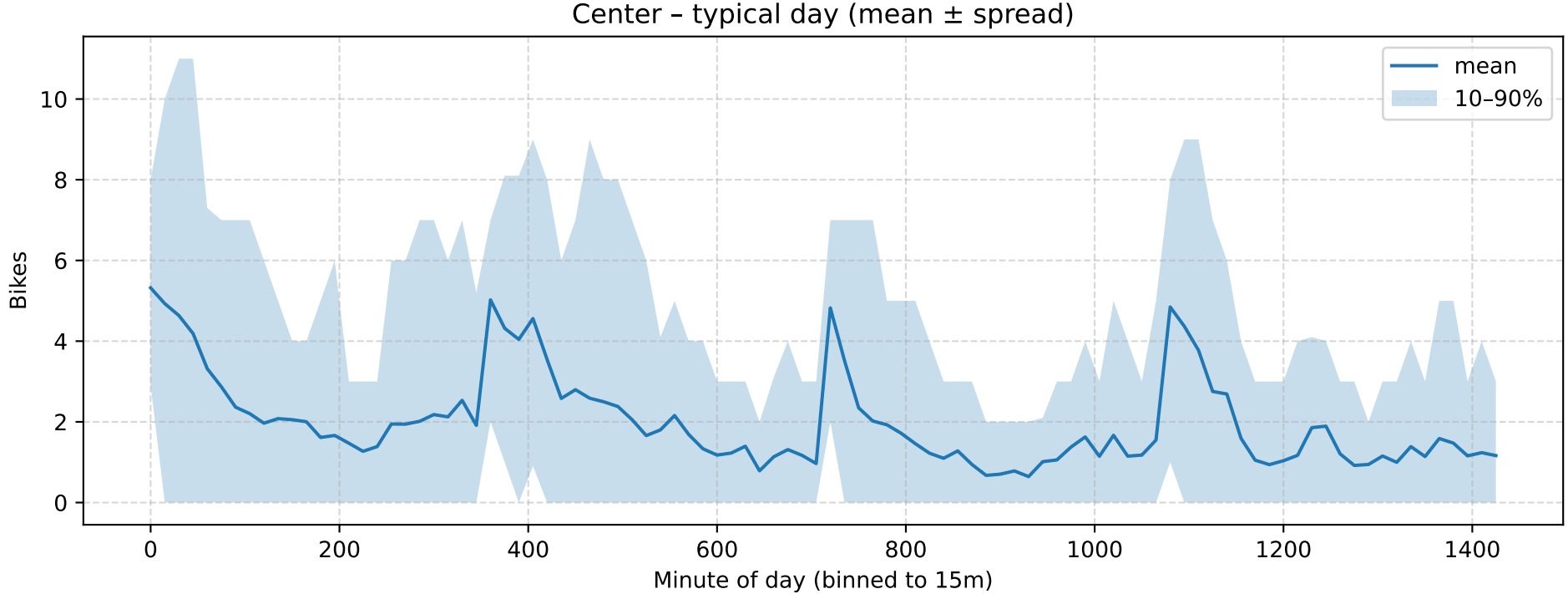

Center – bikes (day vs. time of day)
Day
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
20
40
60
80
Minute of day (binned to 15m)
Bikes
0
2
4
6
8
10
12
14

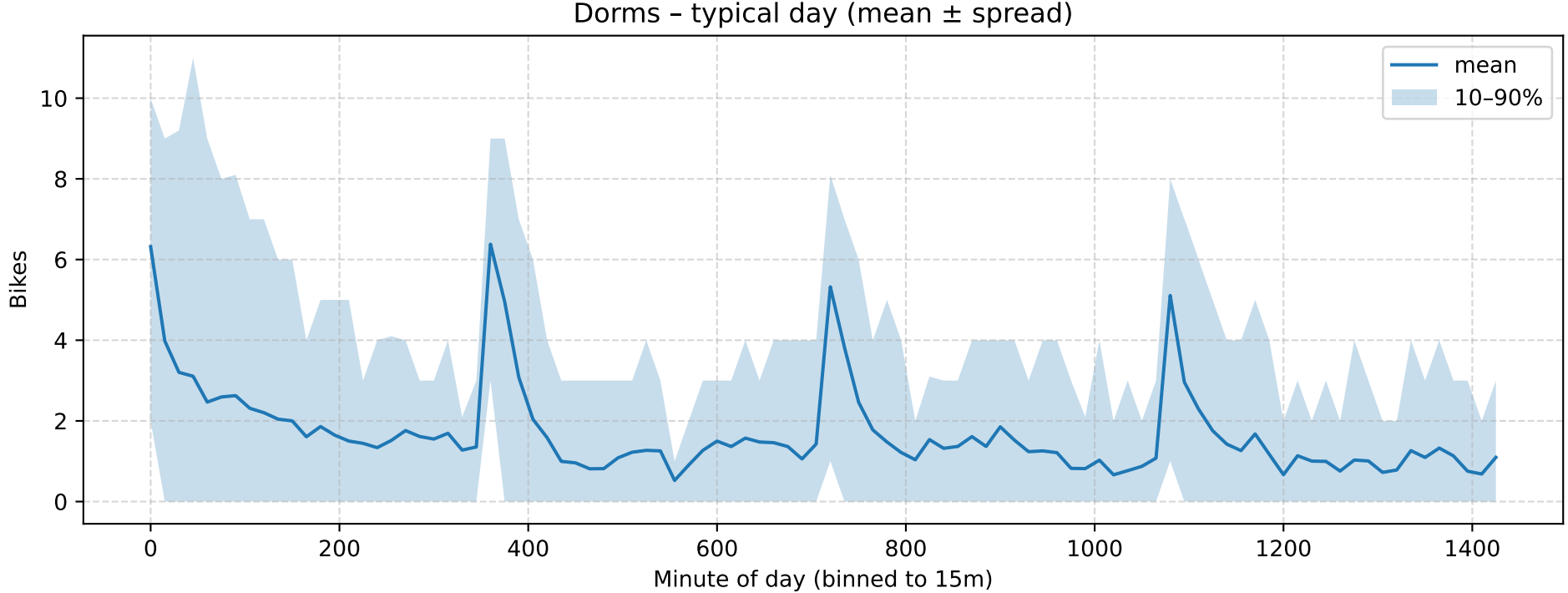

Dorms – bikes (day vs. time of day)
Day
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
20
40
60
80
Minute of day (binned to 15m)
Bikes
0
2
4
6
8
10
12
14

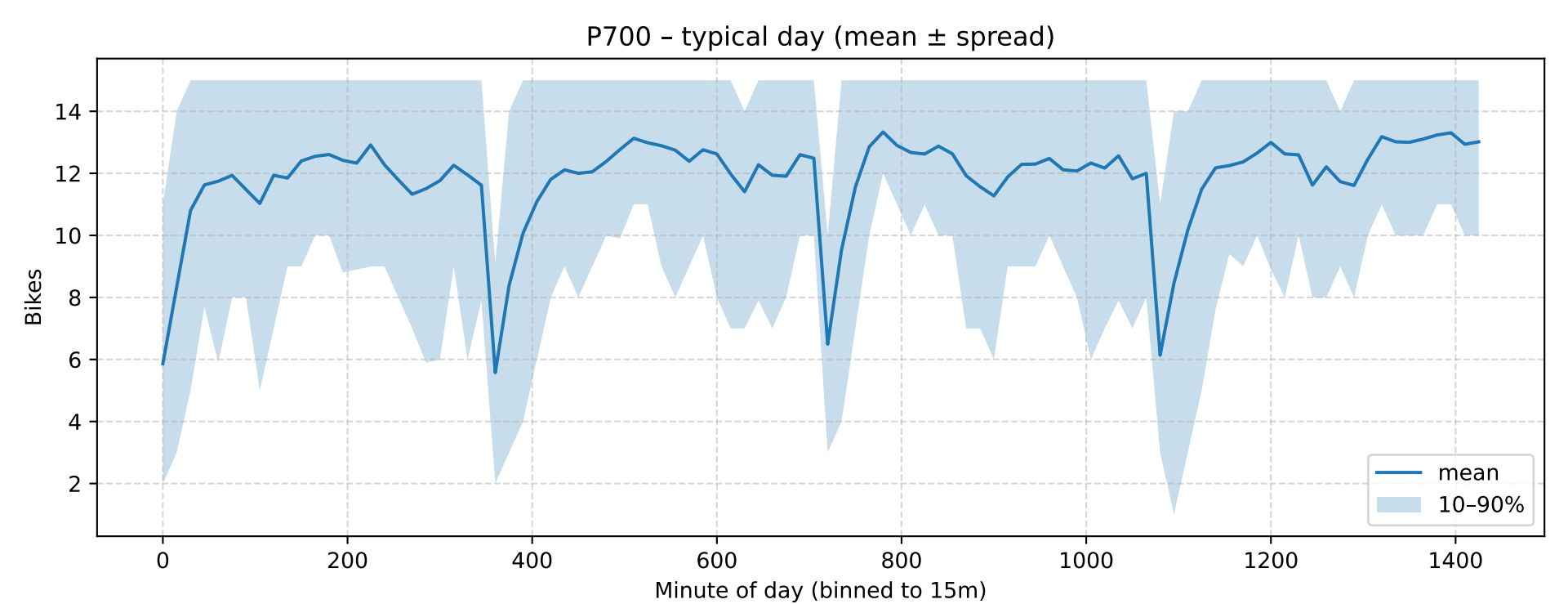

P700 – bikes (day vs. time of day)
Day
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
20
40
60
80
Minute of day (binned to 15m)
Bikes
0
2
4
6
8
10
12
14

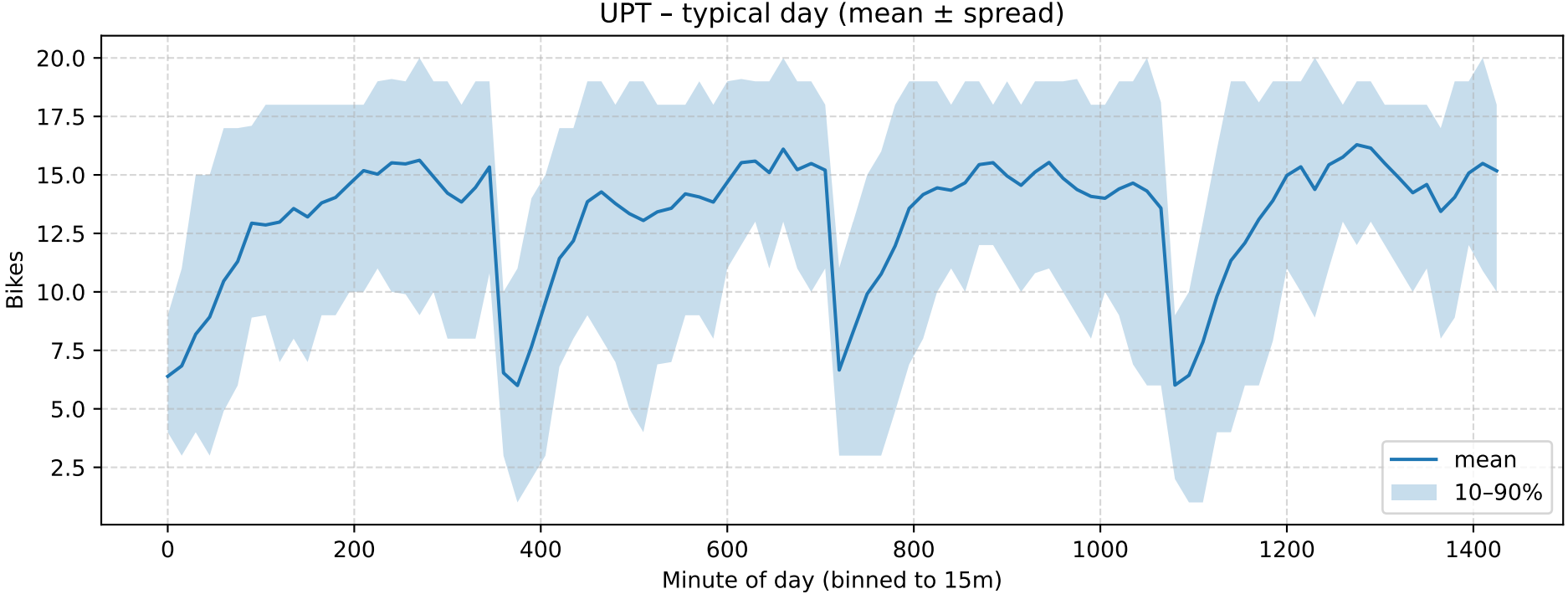

UPT – bikes (day vs. time of day)
Day
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
20
40
60
80
Minute of day (binned to 15m)
Bikes
17.5
15.0
12.5
10.0
7.5
5.0
2.5


Cathedral – typical day (mean ± spread)
mean
10–90%
Bikes
0
2
4
6
8
0
200
400
600
800
1000
1200
1400
Minute of day (binned to 15m)


Center – typical day (mean ± spread)
mean
10–90%
Bikes
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400
Minute of day (binned to 15m)

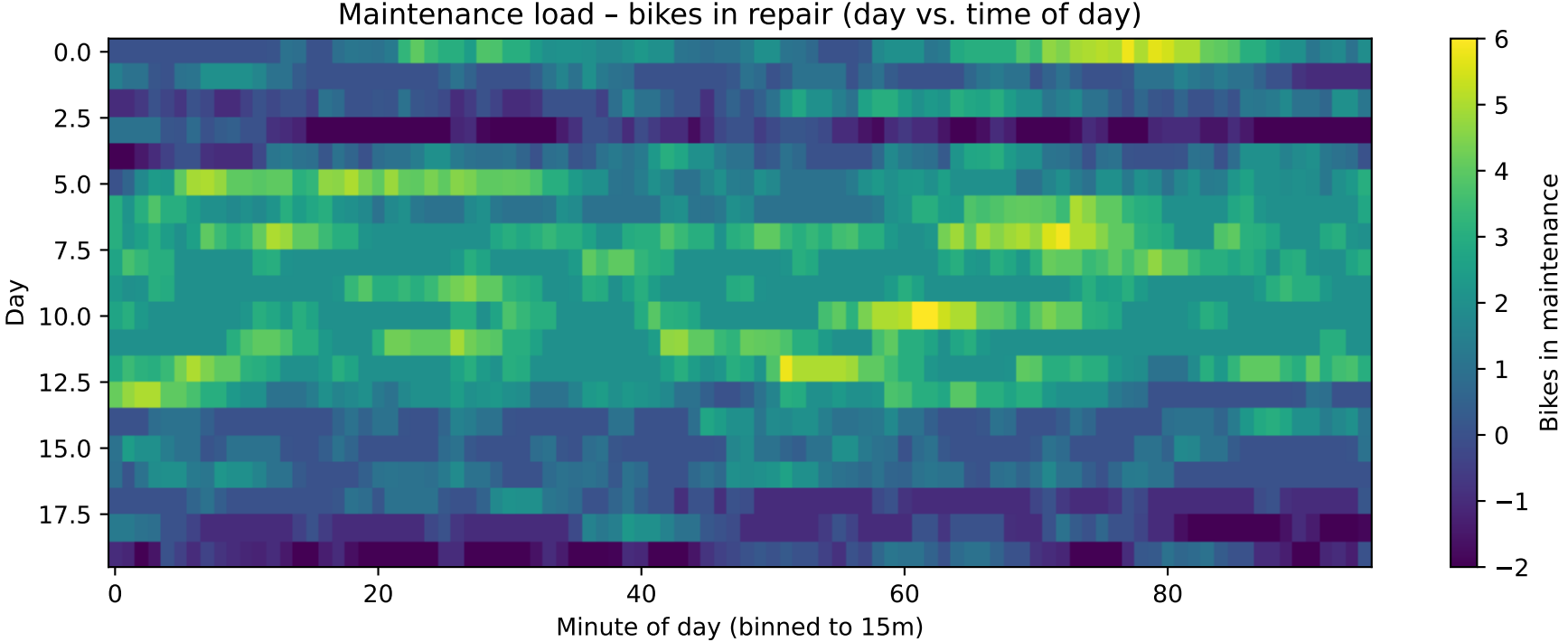

Dorms – typical day (mean ± spread)
mean
10–90%
Bikes
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400
Minute of day (binned to 15m)

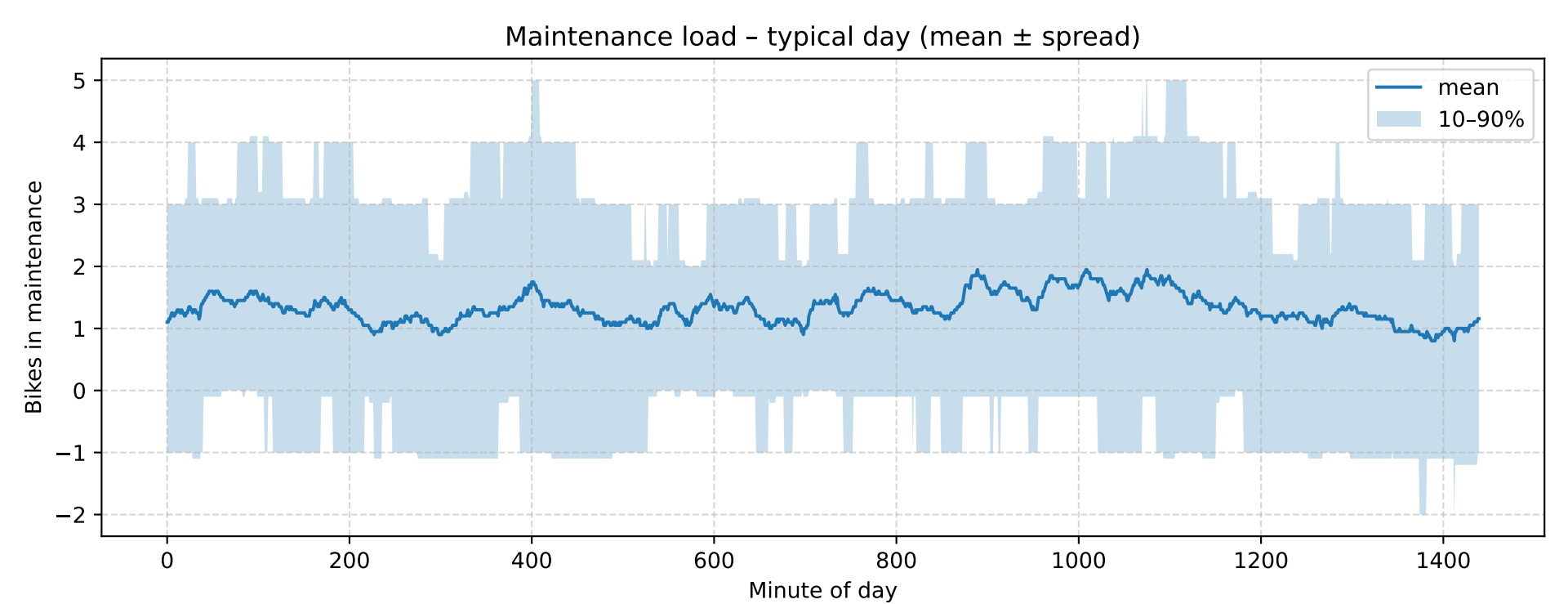

P700 – typical day (mean ± spread)
Bikes
14
12
10
8
6
4
2
0
200
400
600
800
1000
1200
1400
Minute of day (binned to 15m)
mean
10–90%

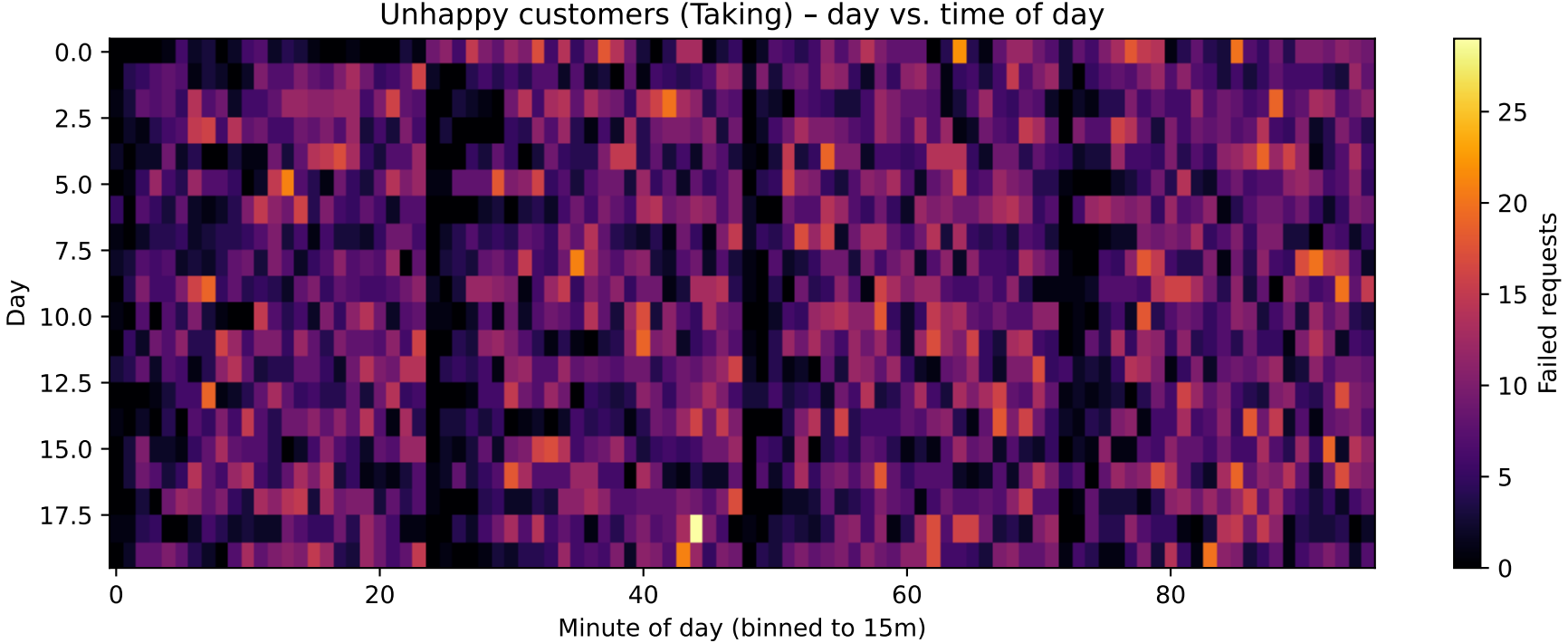

UPT – typical day (mean ± spread)
Bikes
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0
200
400
600
800
1000
1200
1400
Minute of day (binned to 15m)
mean
10–90%

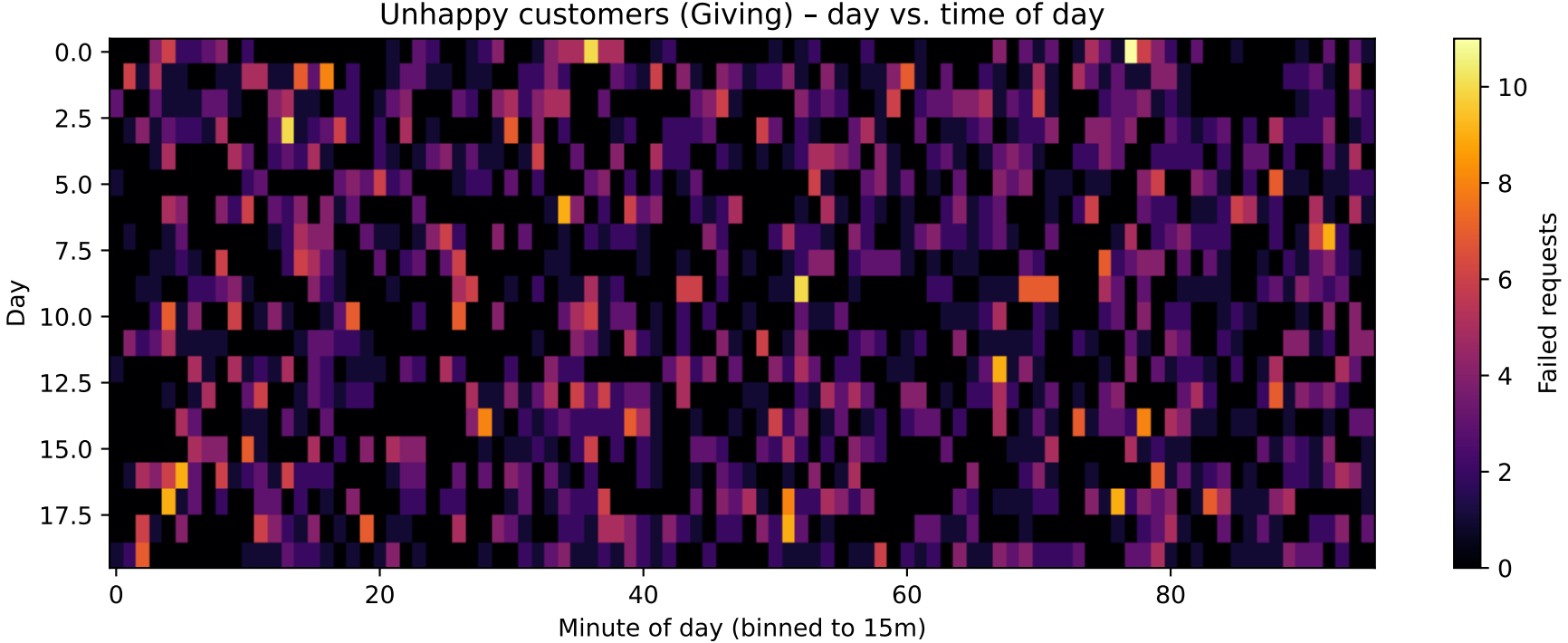
Station KPIs

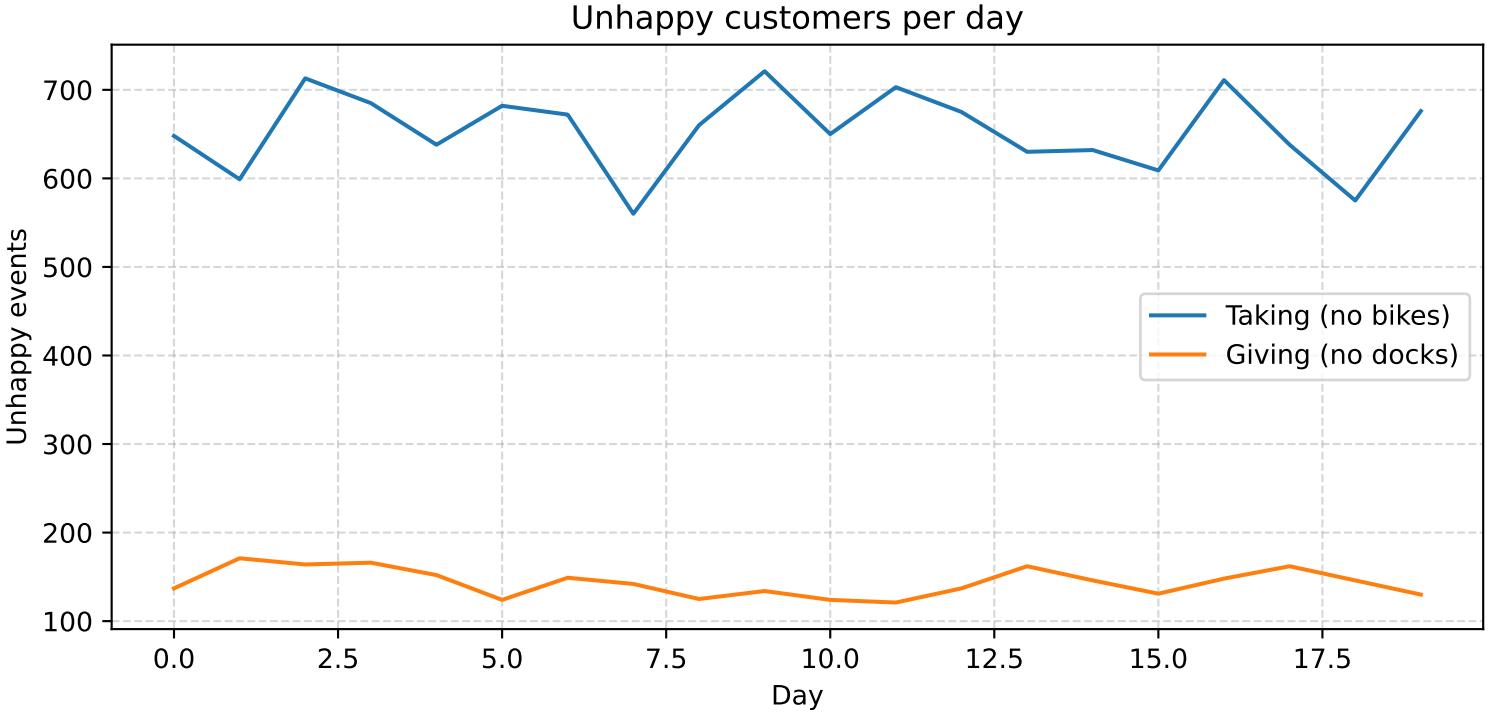
station	avg_bikes	p10_bikes	p90_bikes	stockout_%	overflow_%
Dorms	1.708	0.0	5.0	45.021	0.17
Cathedral	1.995	0.0	6.0	38.49	0.851
Center	2.015	0.0	6.0	38.021	0.333
P700	11.805	7.0	15.0	1.101	20.924
UPT	13.217	6.0	19.0	0.597	3.969

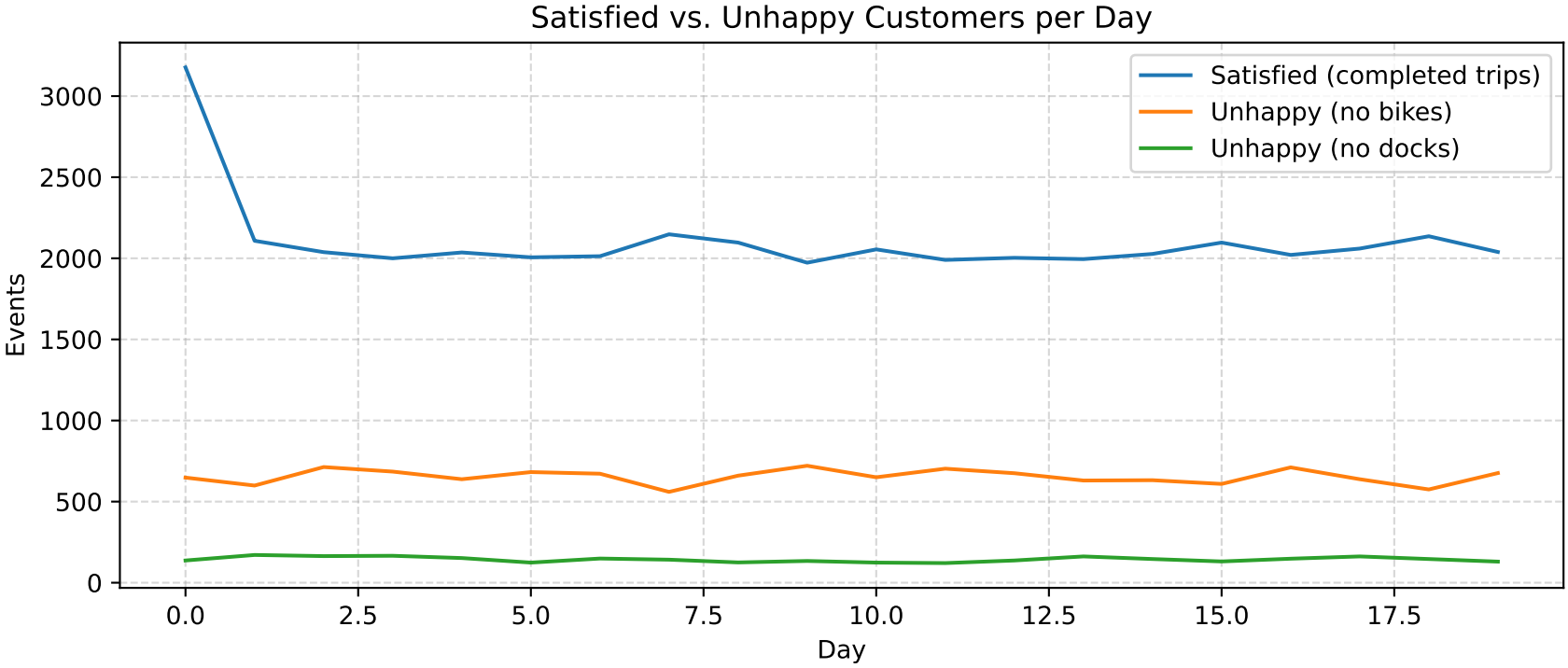

Maintenance load – bikes in repair (day vs. time of day)
Day
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
20
40
60
80
Minute of day (binned to 15m)
Bikes in maintenance
6
5
4
3
2
1
0
−1
−2


Maintenance load – typical day (mean ± spread)
mean
10–90%
Bikes in maintenance
5
4
3
2
1
0
−1
−2
0
200
400
600
800
1000
1200
1400
Minute of day


Unhappy customers (Taking) – day vs. time of day
Day
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
20
40
60
80
Minute of day (binned to 15m)
Failed requests
0
5
10
15
20
25


Unhappy customers (Giving) – day vs. time of day
Day
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
20
40
60
80
Minute of day (binned to 15m)
Failed requests
0
2
4
6
8
10


Unhappy customers per day
Unhappy events
Day
Taking (no bikes)
Giving (no docks)
700
600
500
400
300
200
100
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5


Satisfied vs. Unhappy Customers per Day
Satisfied (completed trips)
Unhappy (no bikes)
Unhappy (no docks)
Events
Day
0
500
1000
1500
2000
2500
3000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5

Total satisfied trips: 42019
Total unhappy attempts: 15948
Satisfaction ratio: 72.49%
Total scrapped bikes: 2